Data-driven Short-term Electric Vehicle Charging Station Demand Forecasting

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***Abstract*—Electric vehicles (EVs) could effectively reduce the dependence of vehicle industry on fossil fuels; however, the complex charging behavior and large power consumption by EVs induce challenges to the operation of power grid. Accurate forecasting of power demand of EV charging station can provide important information for the operation planning of power grid. In this paper, a data-driven framework is proposed for short-term EV charging station demand forecasting. To fully mine the potential of machine learning algorithms in EV charging station demand forecasting, forecasting models based on different machine learning algorithms have been developed including artificial neural network (ANN), support vector regression (SVR), and boosted regression trees (BRT). The performance of a machine learning algorithm highly depends on its hyperpa-rameters. To improve the forecasting performance, a hyperparameter optimization method is further devel-oped based on Rao-l algorithm. To verify the effectiveness of the proposed data-driven framework for EV charging station demand forecasting, extensive computational experiments based on real data from different EV charging stations have been conducted. Experimental results illustrate that the proposed data-driven forecasting models perform much better than the widely considered persistence model while the ANN model and the BRT model outperforms the SVR model.**

***keywords*—Electric Vehicle, Charging Station, Demand Forecasting, Machine Learning, Hyperparameter Optimization.**

I. INTRODUCTION

Affected by the global energy crisis and environmental pollution problems, humans have gradually realized the importance of environmental protection. Driven by the development of a lowcarbon energy structure, electric vehicles (EVs) have gradually become a reliable alternative to internal combustion engine vehicles [1]. According to the International Energy Agency, after a decade of rapid growth, the number of electric vehicles worldwide reached 10 million by the end of 2020, with electric vehicle registrations increasing by 41% in 2020 [2]. However, the largescale use of EVs and electric vehicle charging stations (EVCSs) has brought about a surge in power demand. The surge in power demand has brought huge challenges to the safe and reliable operation of the distribution network [3], [4]. To solve this problem, researchers have proposed solutions including pricing strategies [5] and smart charging [6]. Accurate implementation of the above scheme requires a precise understanding of charging behavior. Therefore, to solve the impact of the increasing charging demand of electric vehicles on the power grid, the prediction of the charging behavior of electric vehicles is of great significance.

The prediction content of charging behavior mainly includes charging duration prediction and charging load prediction [7]. Other prediction contents include the next charging time [8], charging cost [9], etc. Among them, charging load has a direct impact on the power grid and has received the most extensive attention. Load forecasting for electric vehicles can be divided into vehicle-centered forecasting methods and electric vehicle power supply equipment-centered forecasting methods based on different aggregation levels. The first type of method predicts the charging demand of electric vehicles. The charging behavior of electric vehicle users is random and easily affected by schedule interference. The paper [10] is based on the theoretical paradigm of machine thinking, considering the user's historical charging habits and the changing trend of short-term charging demand to predict short-term charging load demand. The paper [11] uses machine learning algorithm to predict the load demand of the primary charging process based on the user's historical charging data and weather forecast data. The paper [12] uses traffic status and car battery status as inputs to the prediction model in the

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prediction process to improve prediction accuracy. The second type of method usually takes charging stations or regions as the research object to predict the charging load in the future. In the paper [13], the author uses Long-Short-Term-Memory (LSTM) to predict the regional electric vehicle load forecast in the next month. The paper [14] uses an integrated algorithm to predict the load of charging stations in the next hour. The integrated algorithm includes artificial neural network (ANN), recurrent neural network (RNN) and long short-term memory network. It can be seen from the above research that most electric vehicle charging load predictions use historical charging data directly as input to the model. Prediction models usually use machine learning algorithms. In addition to the above algorithms, other related research also uses Support Vector Regression (SVM), K-nearest Neighbor (KNN), etc [15], [16]. Machine learning algorithms often have multiple hyperparameters, and the performance of the model largely depends on the values of these parameters, but such parameters are difficult to select effectively.

This paper aims at the problem of short-term load prediction of electric vehicle charging stations. On the one hand, in order to fully tap the potential of machine learning algorithms to solve the load prediction problem of electric vehicle charging stations, prediction models based on different types of machine learning algorithms are established. On the other hand, in order to improve the performance of prediction models based on machine learning algorithms, machine learning hyper-parameter optimization methods based on evolutionary algorithms are studied. In order to verify the effectiveness and efficiency of the prediction model, this paper conducts case analysis based on real electric vehicle charging station observation data sets and compares it with the baseline model.

II. Prediction model

The short-term prediction model framework of electric vehicle charging station load designed in this paper is shown in figure 1. First, the original data is preprocessed, including data cleaning and feature selection, and the data set is divided into a training set and a test set. A prediction model is established based on the training set and machine learning algorithm. In order to improve the prediction accuracy, the hyper-parameters of the machine learning model are further optimized based on evolutionary calculations. Calculate the model prediction error based on the test set to verify the model performance.

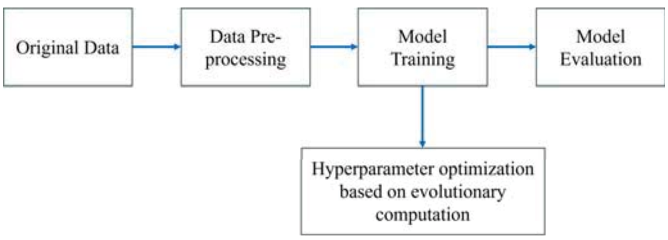

Fig. 1. Charging load forecasting modeling framework

A. Data preprocessing and predictive modeling

This paper conducts research on short-term load prediction of electric vehicle charging stations and uses ACN data [17] for experimental verification. The ACN data set includes recordings of charging sessions at two electric vehicle charging stations at the California Institute of Technology (Caltech) and the Jet Propulsion Laboratory (JPL). This paper uses all records from January 3, 2019 to March 19, 2020. The original data uses a single charging record as metadata. Each record includes the charging start time, power consumption, user information and so on. During the data cleaning process, all charging sessions were converted into time series with a resolution of 1 minute, and the average charging power per hour was calculated as the research object of this article. Figure 2 shows the time series of charging power at JPL electric vehicle charging stations. Figure 2 shows that the load time series of electric vehicle charging stations has obvious daily periodicity. This paper implements one-step prediction of the

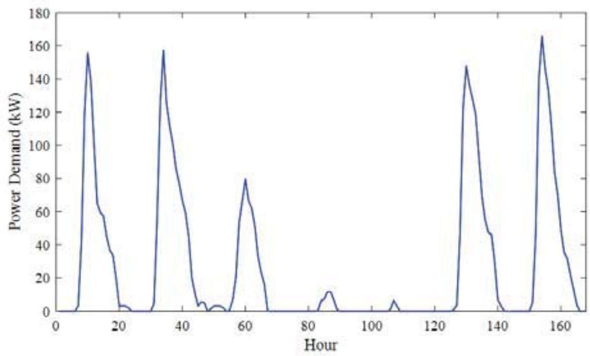

Fig. 2. JPL car charging station load time series

load of electric vehicle charging stations based on the machine learning algorithm and using historical load as characteristics. Let h_t represent the charging load in period t. To realize the prediction of h_t, the input features of the forecast model include the hourly load sequence H_t within 24 hours before the forecast time , which can be define as:

$$H_t = \{h_{t-1}, h_{t-2}, \cdot, h_{t-24}\} \tag{1}$$

The data-driven prediction model can be ex-pressed as:

$$h_t = f(H_t) + e_t \tag{2}$$

In the formula, f represents the machine learning algorithm, e_t represents the model error.

B. Machine learning algorithm

Given a training data set $\mathbb{D} = \{(x_i, y_i)|i = 1, 2, \cdot, N\}$, The machine learning algorithm f establishes the relationship between y and x with the goal of minimizing the difference between $f(x)$ and y. The machine learning algorithms used in this article to establish load prediction for electric vehicle charging stations include Boosted Regression Trees (BRT), Support Vector Regression (SVR) and Artificial Neural Network (ANN).

Enhanced regression tree is an integrated regression analysis algorithm that improves prediction performance by integrating multiple tree-based weak learners. Let $T_m(x, \theta_m)$ represent a tree-based weak learner, where θ_m represents the model parameters, then the enhanced regression tree model integrating M weak learners can be expressed as [18]:

$$f(x) = \sum_{m=1}^{M} T_m(x, \theta_m) \tag{3}$$

The model parameter $\{\theta_m|_{m=1}^{M}\}$ can be obtained by solving equation 4:

$$\{\theta_m|_{m=1}^{M}\} = \arg \min_{\theta'_m|_{m=1}^{M}} \sum_{i=1}^{N} L\left(y_i, \sum_{m=1}^{M} T_m(x, \theta'_m)\right) \tag{4}$$

In the formula, L represents the loss function. This paper applies the square loss function,$L(y, f(x)) = (y - f(x))^2$. Equation 4 is solved using the gradient boosting method in a forward step-wise manner:

$$\theta_m = \arg \min_{\theta} \sum_{i=1}^{N} L\left(y_i, f_{m-1}(x_i) + T_m(x_i, \theta)\right) \tag{5}$$

$$f_m(x) = f_{m-1}(x) + \lambda T(x_m, \theta_m) \tag{6}$$

In the formula, The hyper-parameter λ represents the learning rate. $T_m(x, \theta_m)$ is obtained by fitting the current residual $y - f_{m-1}(x)$.

Support vector regression is a regression model based on kernel function and structural risk minimization. This article considers the ϵ-SVM model [19], which is shown in equation 7:

$$f(x) = w\varphi(x) + b \tag{7}$$

In the formula, w and b are the model parameter, $\varphi(x)$ maps the input space into a high-dimensional feature space. w and b can be obtained by introducing slack variables to solve equation 8:

$$\min \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \xi + \xi^* s.t. \begin{cases} y_i - w\varphi(x_i) - b \leq \epsilon + \xi_i \\ w\varphi(x_i) + b - y_i \leq \epsilon + \xi_i \\ \xi_i, \xi_i^* \geq 0 \end{cases} \tag{8}$$

In the formula, C is the penalty factor, ϵ is the in-sensitivity coefficient, ξ_i and ξ_i^* are slack variables. To solve the equation 8, its Lagrangian dual form can be introduced, and finally a prediction model shown in equation 9 can be obtained:

$$f(x) = \sum_{i=1}^{N} \left(\alpha_i - \alpha_i^*\right) K(x_i, x) + b \tag{9}$$

In the formula, α_i and α_i^* are the Lagrange multiplier, $K(x_i, x) = \varphi(x_i)\varphi(x)$ is the kernel function. This paper uses the radial basis kernel function shown in equation 10:

$$K\left(x_i, x_j\right) = exp\left(\frac{\|x_i - x_j\|^2}{-2\sigma^2}\right) \tag{10}$$

In the formula, σ is the kernel function control coefficient.

Artificial neural network is the most commonly used regression analysis model because of its powerful fitting ability. This paper uses a Multi-layer Perception (MLP) with 1 hidden layer to establish a load prediction model for electric vehicle charging stations. The performance of multi-layer perceptron depends on the number of hidden layer neurons, activation function and network parameter optimization method. In the input layer, the model input features are directly passed to the hidden layer. The hidden layer uses tanh activation function, as shown in equation 11. The output layer uses a linear activation function. Levenberg-Marquardt optimization algorithm [20] is used for neuron weight optimization.

$$\tan sig(u) = \frac{2}{1 - exp(-2u)} - 1 \tag{11}$$

C. Hyper-parameter optimization method of machine learning algorithm

The performance of machine learning algorithms is highly dependent on the selection of hyper-parameters. This article uses the Rao-1 algorithm based on cluster intelligence ideas for hyper-parameter optimization of machine learning algorithms. The hyper-parameters to be optimized for each machine learning algorithm are shown in Table 1.

TABLE I
CALTECH CHARGING STATION PREDICTION RESULTS

Data mining algorithm	Hyper-parameters
Boosted Regression Tree (BRT)	Number of weak learners (M)
	Learning rate (λ)
	The maximum number of splits of the weak learner (p)
Support Vector Regression (SVR)	Insensitivity coefficient (ϵ)
	Penalty coefficient (C)
	Kernel function control coefficient σ
Artificial Neural Network (ANN)	Number of hidden layer neurons (Q)

Evolutionary algorithms such as swarm algo-rithms and genetic algorithms have been widely used in engineering optimization problems and have achieved good application results. However, the performance of the above evolutionary algorithms depends on the hyperparameters of the algorithm itself, which limits the use of the above optimiza-tion algorithms to a certain extent. Compared with the example group algorithm and genetic algorithm, the biggest feature of the Rao-1 algorithm is that the algorithm is simple and the algorithm itself has no parameters.

Let φ be the set of hyper-parameters of a certain machine learning algorithm, then the selection of hyper-parameters of the machine learning algorithm can be expressed as the optimization problem shown in equation 12:

$$\min_{\varphi} L\left(y, f(x|\varphi)\right) \tag{12}$$

In order to seek the optimal φ, Rao-1 first initializes the candidate population, trains the corresponding machine learning model for each initial solution and calculates L, and update the population based on L. Obtain the optimal solution through continuous iterative updates. Let $\varphi_{i,j,k}$ represents the k element of j candidate solutions in iteration i, The iterative update method of Rao-1 algorithm is shown in Equation 13:

$$\varphi'_{i,j,k} = \varphi_{i,j,k} + r_{i,k}(\varphi_{i,best,k} - \varphi_{i,worst,k}) \tag{13}$$

In the formula, $\varphi_{i,best,k}$ and $\varphi_{i,worst,k}$ represents the k element of the best candidate solution and the worst candidate solution of the i generation respectively.$r_{i,k}$ are random values between [0,1]. The hyper-parameter optimization process based on the Rao-1 algorithm is shown in figure 3.

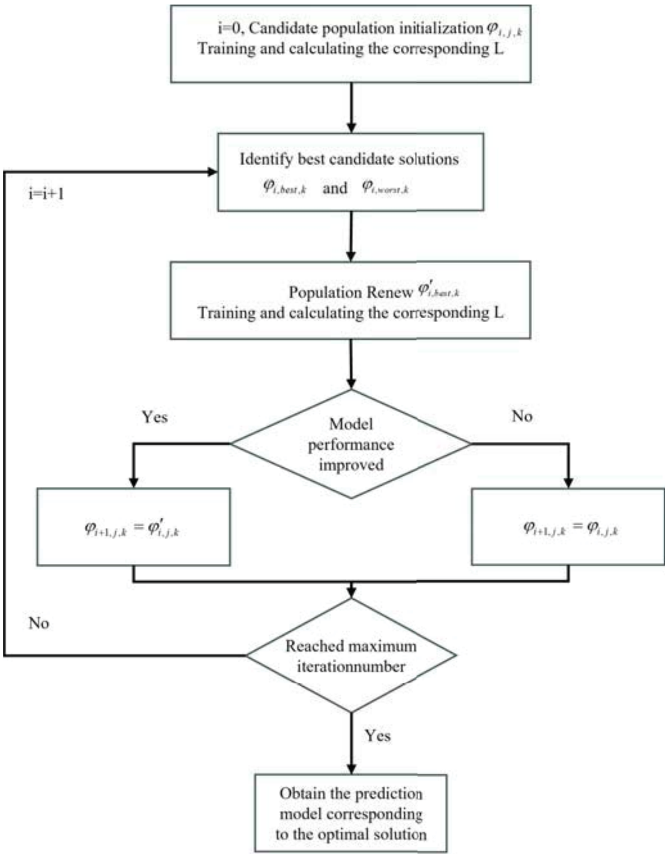

Fig. 3. Hyper-parameter optimization method of machine learning algorithm based on Rao-1

III. Experimental results and analysis

A. Model evaluation indicators

To compare the prediction performance of each model, the Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Coefficient of Determi-nation (R2) are selected as evaluation indicators, such as (14)-(16) as shown.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{14}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{15}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{16}$$

In the formula, y_i and $\hat{y}_i$ represent the observed values and predicted values respectively, and $\bar{y}$ is the average of the observed values.

B. Results and analysis

The data set is divided into a training set and a test set in chronological order at a ratio of 7:3, that is, a total of 310 days of load data from January 3, 2019 to November 8, 2019 is the training set, and from November 9, 2019 to The load data for a total of 132 days on March 19, 2020 is the test set. To fully reflect the performance of machine learning methods in the prediction task of electric vehicle charging stations, this paper combines the persistence model (PER), ,$\hat{h}_t = h_{t-1}$ is set as the baseline model. The persistence model is simple and performs better in short-term prediction tasks, so the persistence model is often used as a comparison model for short-term prediction.

TABLE II
JPL charging station prediction results

	MAE(kW)	RMSE(kW)	R^2
PER	8.28	16.34	0.855
BRT	4.15	7.30	0.971
SVM	5.30	10.39	0.941
ANN	4.19	6.93	0.974

TABLE III
Caltech charging station prediction results

	MAE(kW)	RMSE(kW)	R^2
PER	3.47	6.69	0.745
BRT	2.83	4.76	0.871
SVM	3.14	5.76	0.811
ANN	3.00	4.50	0.885

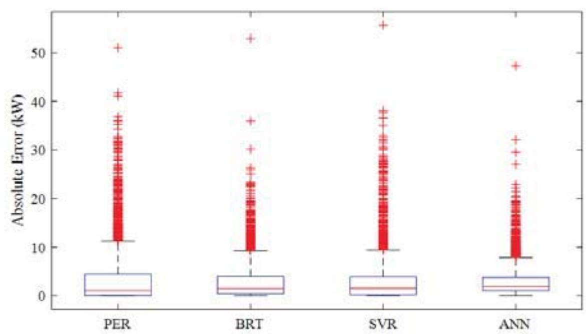

Fig. 4. Forecast absolute error box plot of JPL

The data-driven short-term prediction results of electric vehicle charging station load are shown in Tables 2 and 3. As can be seen from the table, the performance of the prediction model based on ma-chine learning algorithms is significantly better than the persistence model, but there are

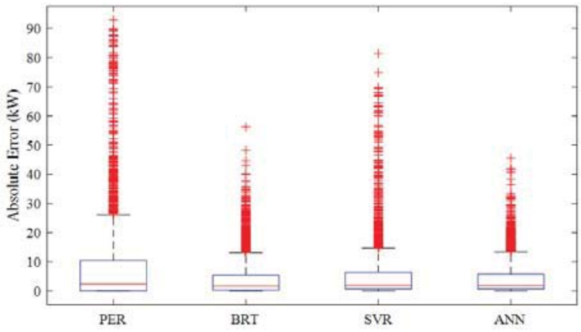

Fig. 5. Forecast absolute error box plot of Caltech

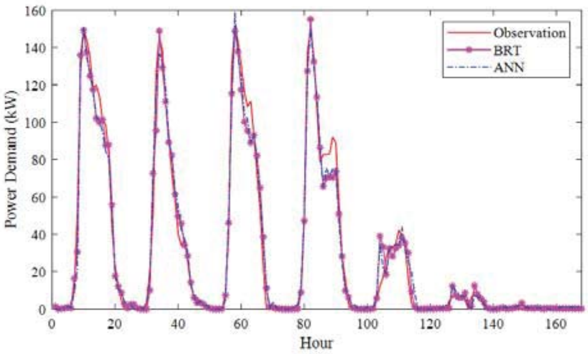

Fig. 6. Data mining algorithm prediction results of JPL

certain differences in the performance of each data mining model. BRT and ANN are significantly better than SVR. For MAE, the BRT model is better than the ANN model; but for RMSE and R2, the ANN model is better than the BRT model.

Figure 4 and 5 show the box plot of the prediction errors of each model. It can be seen from the figure that the BRT model and ANN model are significantly better than the PER model and SVR model. Compared with the ANN model, the BRT model is more likely to produce prediction results with larger absolute errors. The calculations of RMSE and R2 are based on squared errors and are sensitive to larger errors. Therefore, although the BRT model is superior in MAE index, its RMSE and R2 are not as good as the ANN model.

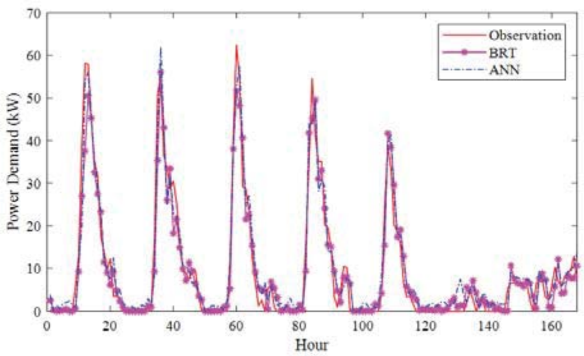

Fig. 7. Data mining algorithm prediction results of Caltech

Figure 6 and 7 shows the prediction results of the BRT model and the ANN model. It can be seen from the figure that the BRT model and ANN model can better track the changes in load, thereby providing more accurate prediction results.

IV. CONCLUSION

This paper designs a data-driven prediction model framework for the short-term prediction problem of electric vehicle charging station load. On the one hand, in order to fully tap the potential of machine learning algorithms in the load forecasting task of electric vehicle charging stations, three different types of machine learning algorithms were selected, including enhanced regression trees, sup-port vector regression and artificial neural networks. On the other hand, in view of the fact that the performance of machine learning algorithms is highly dependent on algorithm hyperparameters, a machine learning algorithm hyperparameter opti-mization method based on Rao-1 was designed. This paper uses public and real electric vehicle charging station observation data to verify the effectiveness and efficiency of the prediction model. Experimental results show that compared with persistence models, data mining models based on machine learning algorithms can significantly improve pre-diction accuracy. There are certain differences in the performance of each data mining model. The pre-diction performance of enhanced regression tree and artificial neural network is better than that of sup-port vector regression.

V. ACKNOWLEDGEMENT

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